Supporting Information: title to be chosen

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Methods

Baselines

> **Plan**
>
> - `randomwalk` and `randomwalknowcast` in full.
> - Key references: (Johnson et al. 2025).

Foundation model setup and fine-tuning

Plan

- How Chronos-2 and LoRA work, in brief, with schematics.
- Foundation model setup and fine-tuning settings.
- Key references: (Ansari et al. 2025; Hu et al. 2021).

Synthetic data generation

Plan

- Synthetic data generation in full, with an overview of the epinowcast model used.
- Key references: (Abbott et al. 2026; Lison et al. 2024; Park et al. 2024; Abbott 2026).

Scoring

Plan

- Scoring, and its check against scoringutils.
- Key references: (Bosse et al. 2022; Jordan, Krüger, and Lerch 2019; Zanetta and Allen 2024).

Implementation

Plan

- The tools we used, each cited: Python, PyTorch on the laptop GPU, chronos-forecasting, PEFT for LoRA, R and epinowcast for simulation, and scoringrules for scoring, checked against scoringutils.
- The evaluation runs as a Snakemake workflow, so every result can be reproduced from the raw data.

Results

> **Plan**
>
> - Score breakdowns by location and horizon for each case study.
> - WIS over time for each case study.
> - Held-out synthetic scores by number of fine-tuning outbreaks.

Held-out synthetic data

German nowcast hub

Ebola hub submissions

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